ST WEEK OF

LOPMENT A

RTILIZATIO

LEARNING OBJECTIVES

At the end of the lecture, the students should be able to :

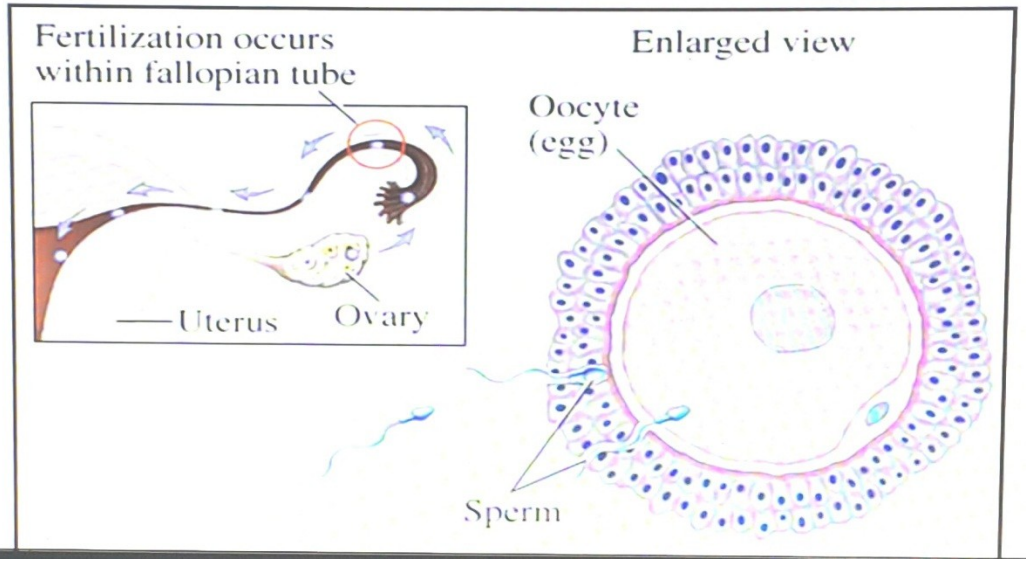
- Discuss the formation of zygote.
- Correlate the transport of zygote from ampulla of fallopian tube to the uterine cavity and cleavage.
- Explain the formation of blastocyst.

FERTILIZATION

Definition:

The **union** of **male gamete(sperm)** and **female gamete (oocyte)** gamete to form a zygote.

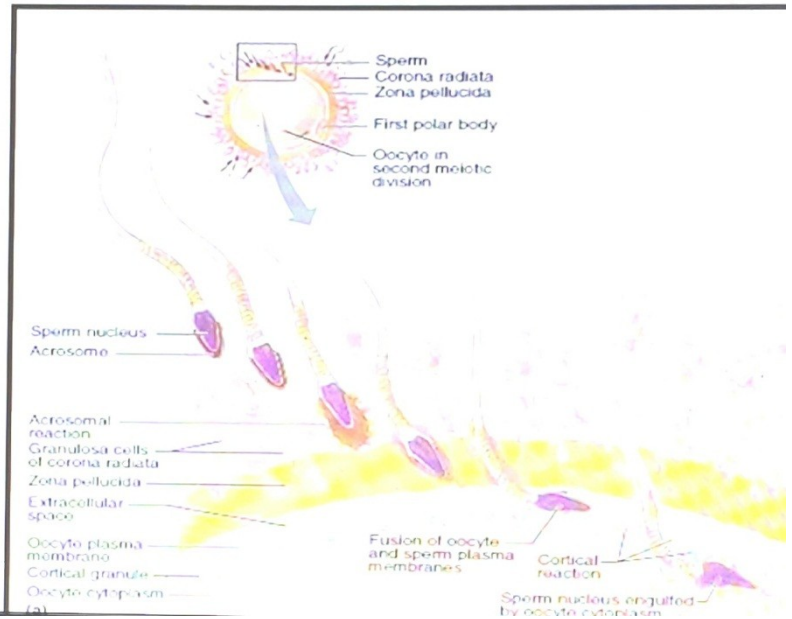
The process takes approximately 24 hours.



FERTILIZATION

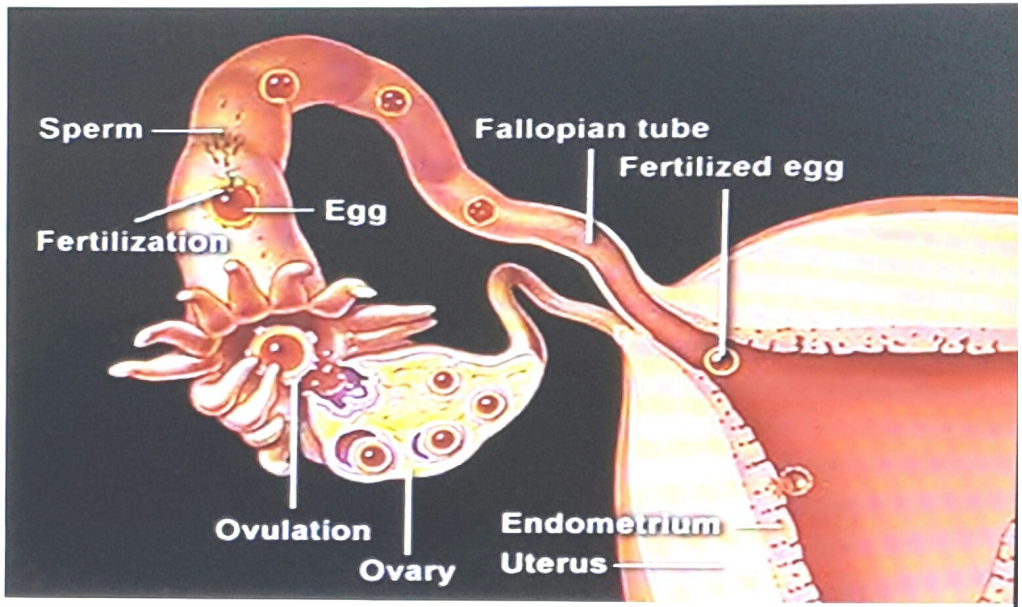
Begins with a **contact** between the sperm & the ovum.

Ends up with **intermingling** of the maternal and paternal chromosomes.



SITE OF FERTILIZATION

- Usual site is the **ampulla of the uterine tube.**
- Ampulla is the longest and widest part.
- **Chemical signals (attractants)** , secreted by oocyte and surrounding follicular cells, guide capacitated sperms (sperm chemotaxis) to the oocyte.

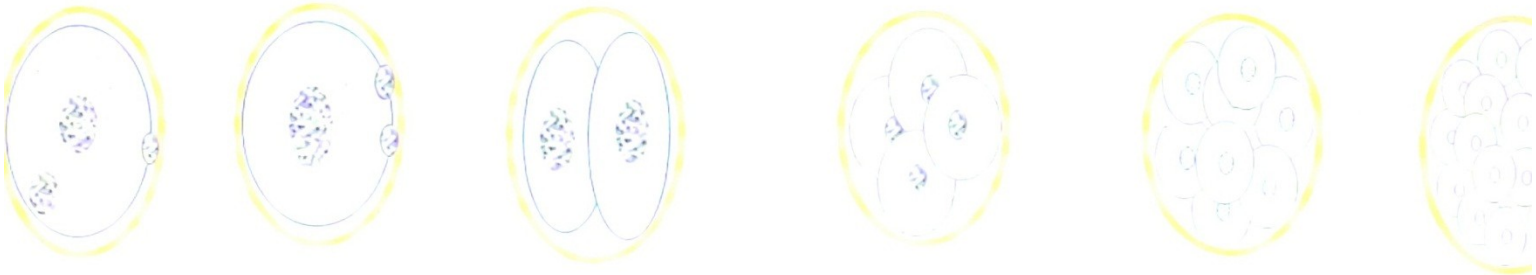


RESULTS OF FERTILIZATION

- Restores the normal diploid number of chromosomes in the zygote (46).
- Variation of human species through mingling of maternal and paternal chromosomes.
- Determines the chromosomal sex of the embryo.
- Initiates cleavage (cell division) of the zygote.

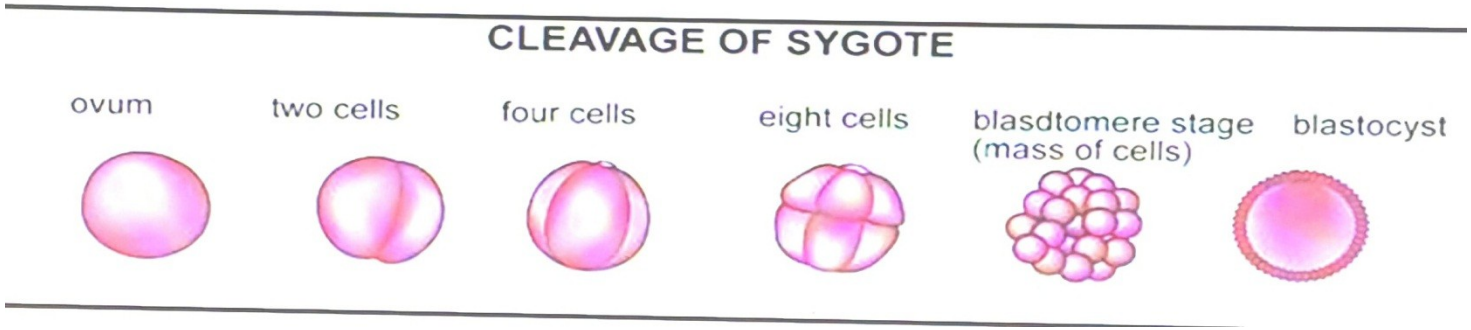
CLEAVAGE OF ZYGOTE

- Consists of repeated mitotic divisions of the zygote.
- Rapid increase in the number of the cells.
- These embryonic cells are called **Blastomeres.**
- **Blastomeres become smaller with each successive division.**
- Cleavage Normally occurs in the **uterine tube.**



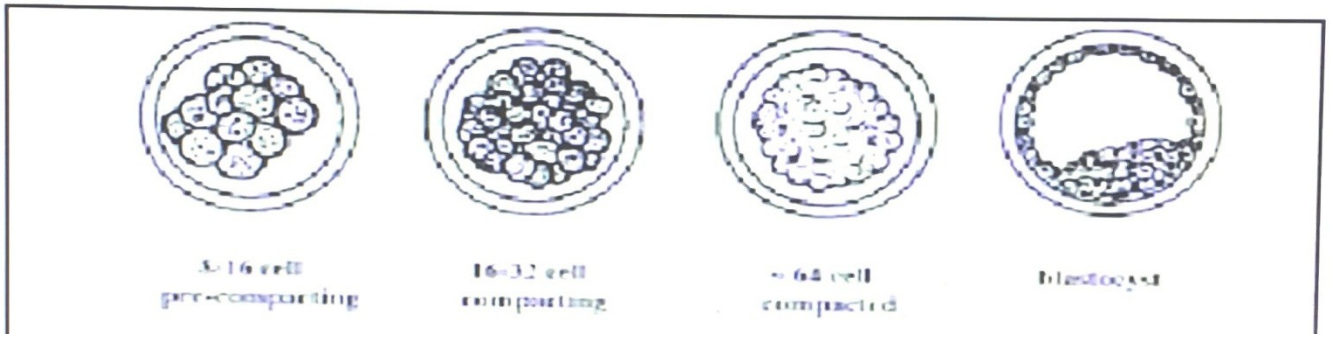
CLEAVAGE OF ZYGOTE

- It begins about **30** hours after fertilization.
- Zygote divides into 2, then 4, then 8, then 16 cells.
- Zygote lies within the thick **zona pellucida** during cleavage.
- Zygote migrates in the uterine tube from its lateral end to its medial end.



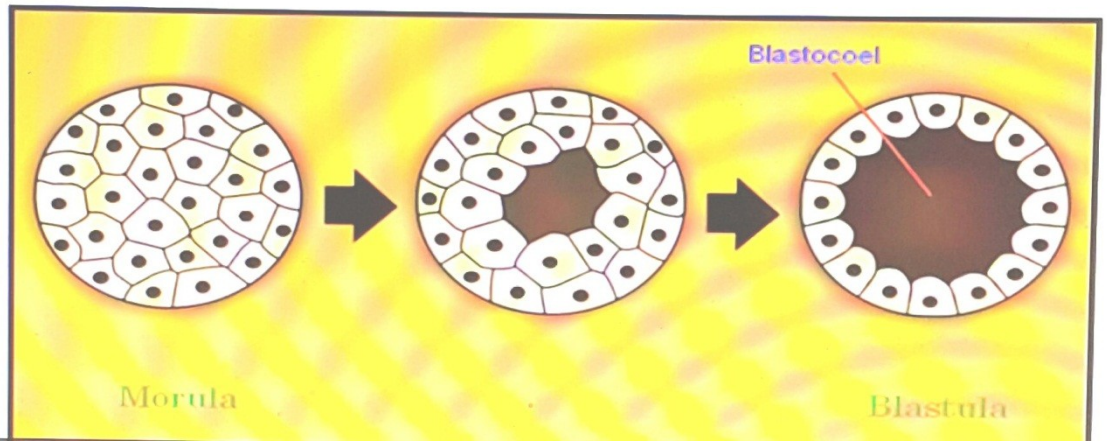
COMPACTION

- After eight cell stage, **Blastomeres** maximize their contact with each other and each cell is attached with other cells by the tight junctions. this process is called as Compaction.
- **This Compaction** is Mediated by cell surface **adhesion glycoproteins**.
- The inner cells of the morula are called inner cell mass later these cells will form the tissue of the embro proper.



MORULA

- When there are 12-32 blastomeres the developing human is called **MORULA.**
- Spherical **Morula is** formed about **3** days after fertilization.
- The outer cells of the morula are called outer cell mass or Trophoblast which later forms PLACENTA.



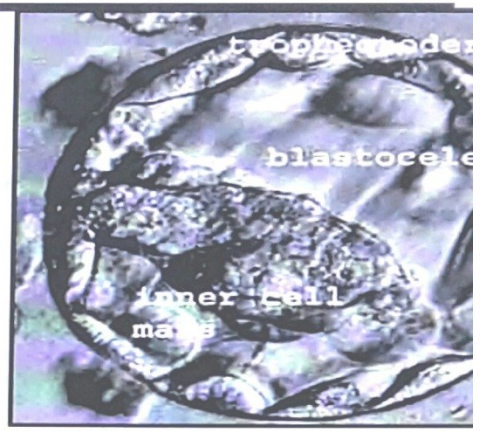
FORMATION OF BLASTOCYST

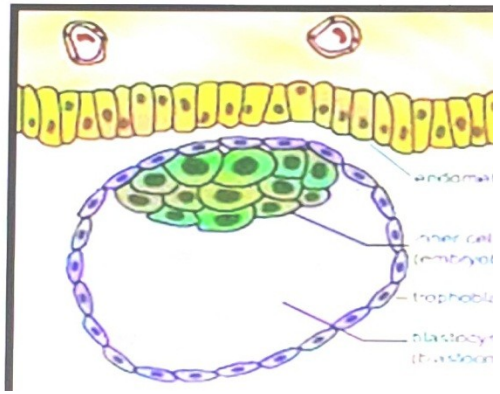
- **The Morula** reaches the uterine cavity by the 4th day after fertilization, & remains free for one or two days.

Fluid passes from uterine cavity to the intercellular spaces of the inner cell mass of the Morula through zona pallucida.

The spaces containing fluid join to form a single large space called BLASTOCYST cavity or blastocoele.

- Now the Morula is called ***Blastocyst.***



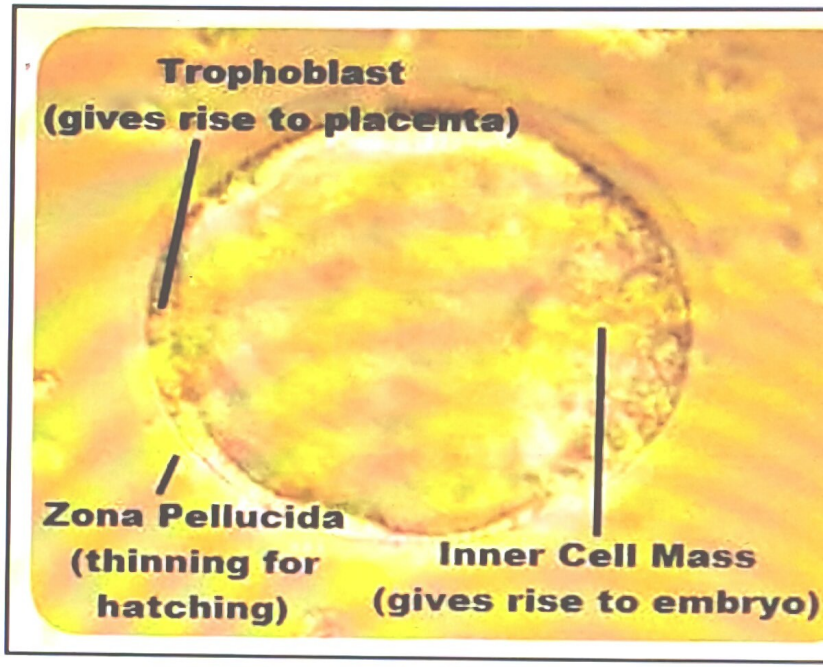


BLASTOCYST

Trophoblast:

- A thin ,outer cell layer ,which gives rise to embryonic part of placenta.
  - The cytotrophoblast is the inner layer of the trophoblast .
  - Syncytiotrophoblasts are multinucleated cells found in the placenta of embryos.
  - They are the outer syncytial layer of the trophoblasts.

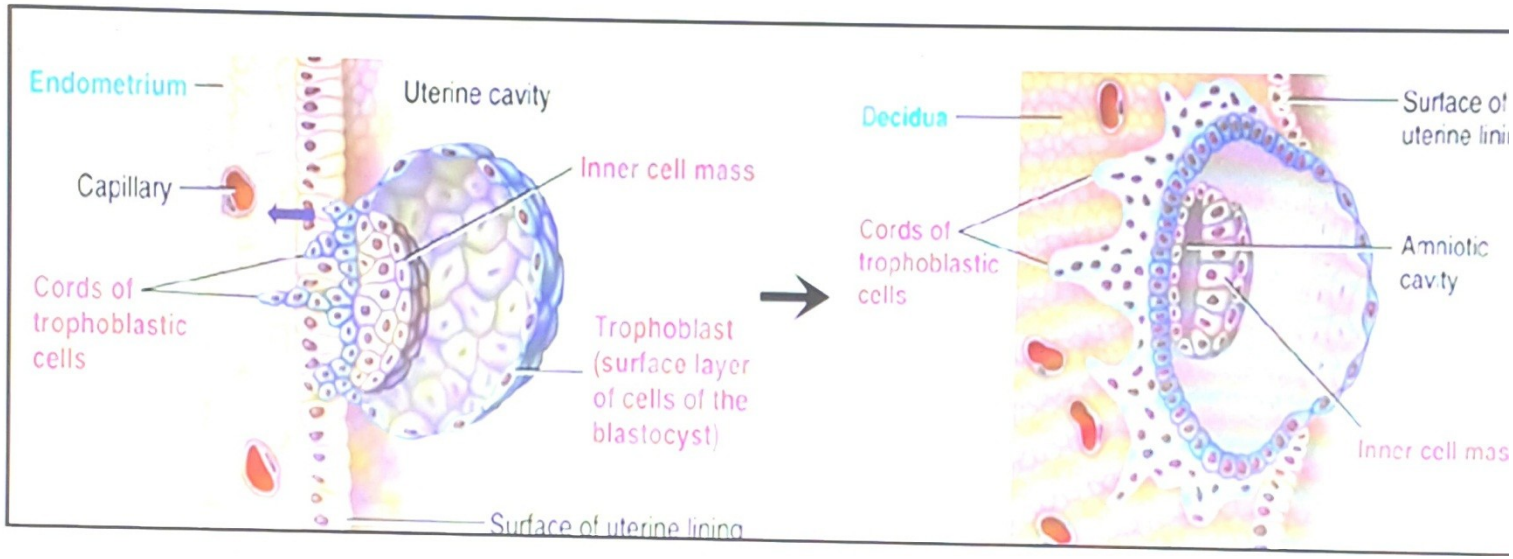
- Embryoblast: Group of centrally located blastomeres, the inner cell mass, give rise to embryo or embryoblast.



IMPLANTATION

Definition :

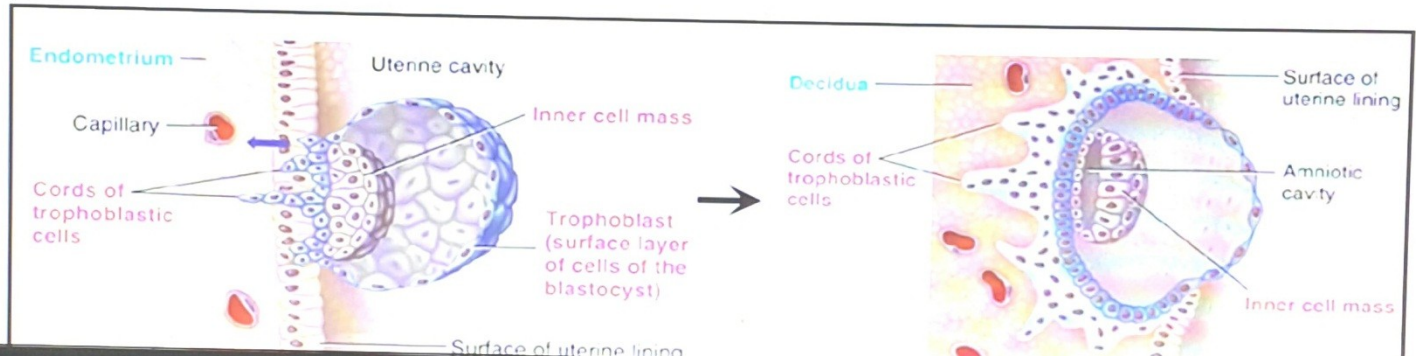
- It is the process by which the **Blastocyst** penetrates the **superficial** (Compact) layer of the endometrium of the uterus.



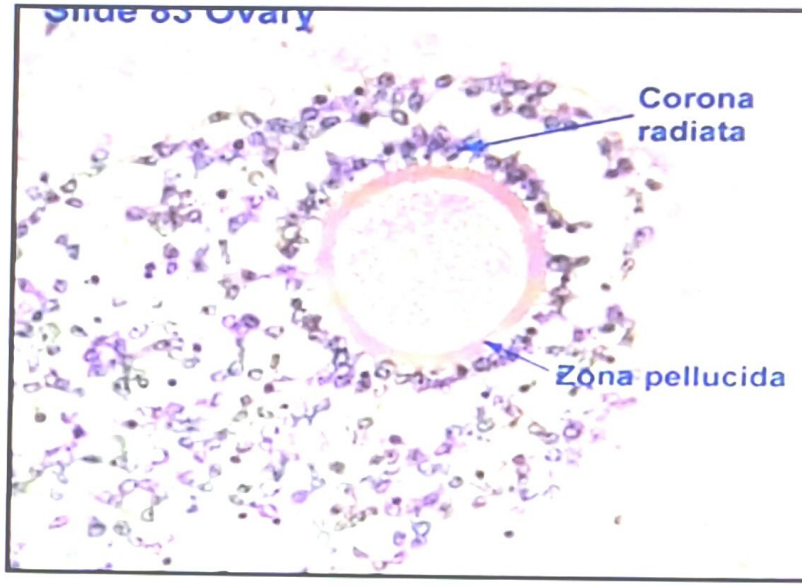
IMPLANTATION

Site:

- The normal site of ***implantation*** is the **posterior and anterior wall of body of the uterus near the fundus**.
- ***Time:***
- It **begins** about the **6th day** after fertilization or nearly end of the first week.
- It is **completed** by the 11th or 12th day.



- By the **5th day** the Zona pellucida degenerates.
- Blastocyst begins **implantation** by the **6th** day.
- Trophoblast cells penetrate the epithelium of the endometrium.
- Penetration results from proteolytic enzymes (eg.COX-2) produced by the **trophoblast**.



- **Zona pellucida** degenerates & disappears by the **5th** day to allows the **blastocyst** to increase in size and penetrates the endometrium.
- ***The embryoblast*** projects into the blastocystic cavity, while **the trophoblast** forms the wall of the blastocyst.
- **By 6th day** the blastocyst adheres to the endometrial epithelium.
- **By 7th day,** T*rophoblast* differentiated into 2 layers:

Cytotrophoblast, inner layer, mononucleated mitotically active cells.

Syncytiotrophoblast (outer multinucleated mass, with *indistinct* cell boundary.

By 8th day the blastocyst is superficially embedded in the compact layer of the endometrium (by the end of 1st week,the blastocyst is superficially implanted in endometrium).

- ***Endometrial cells*** undergo **apoptosis** (programmed cell death) to facilitates invasion of endometrium by the Syncytiotrophoblast.

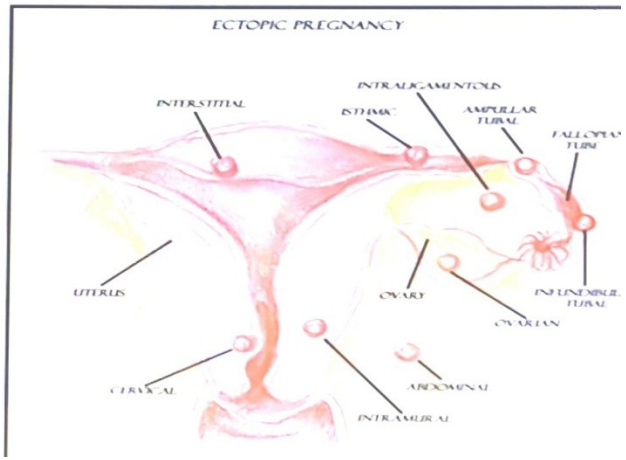
- **Syncytiotrophoblast** engulf these degenerated cells for nutrition of the embryo.

- ***Implantation***
  can be detected by:
  1- Ultrasonography.
  2- hCG (human chorionic gonadotrophin which is secreted by the Syncytiotrophoblast) about **the end of 2nd week**

ECTOPIC PREGNANCY

- **It means <u>implantation outside the uterus.</u>**
- <u>95 to 97%</u> of ectopic pregnancies occurs in the <u>uterine tube.</u>
- Most are in the <u>ampulla & isthmus.</u>
- **<u>Placenta previa :</u>**
- Implantation occurs in <u>the lower uterine segment.</u>

- **<u>Ectopic Pregnancy</u>:**
- **1- Placenta Previa.**
- **2- Tubal.**
- **3- Ovarian.**
- **4- Abdominal.**
- **5- Pelvic.**
- **6- Cervical.**



REFERENCES

- Keith L.Moore Developing human 8th Edition Chapter 2 pages 30-40.